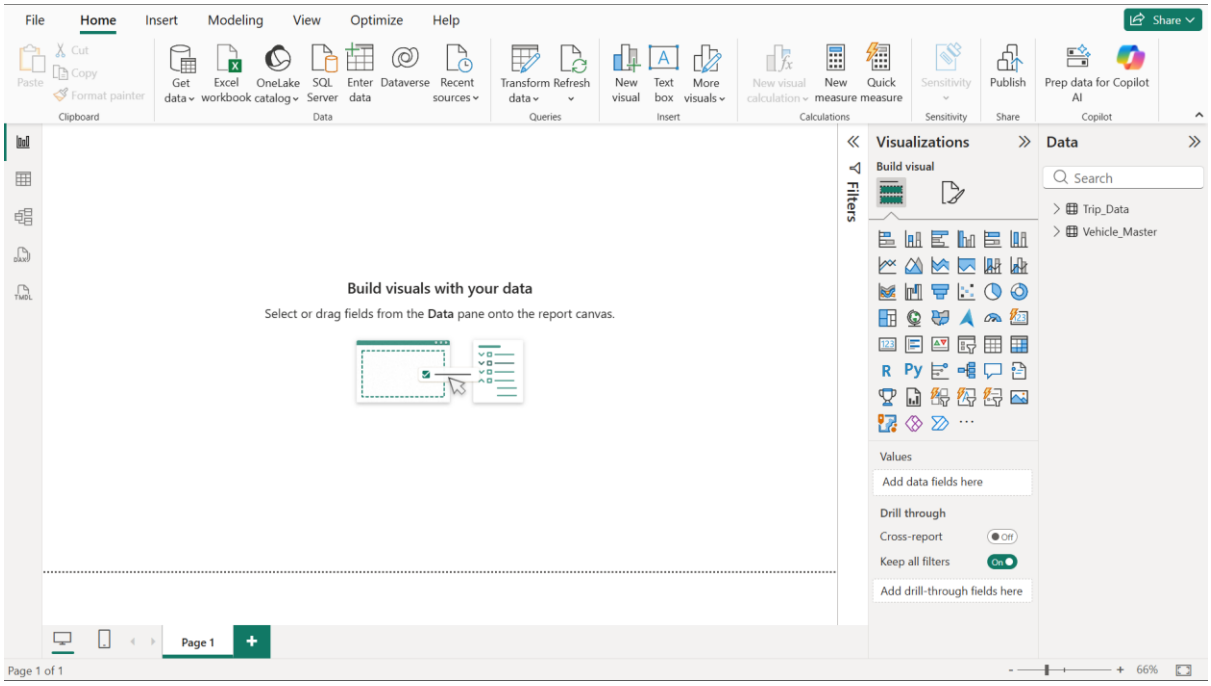
Module End Assignment

The objective of this assignment is to develop a Transport Operations Dashboard using Power BI to analyze delivery performance, fuel efficiency, and operational costs through data cleaning, modeling, DAX measures, and AI-driven visualizations.

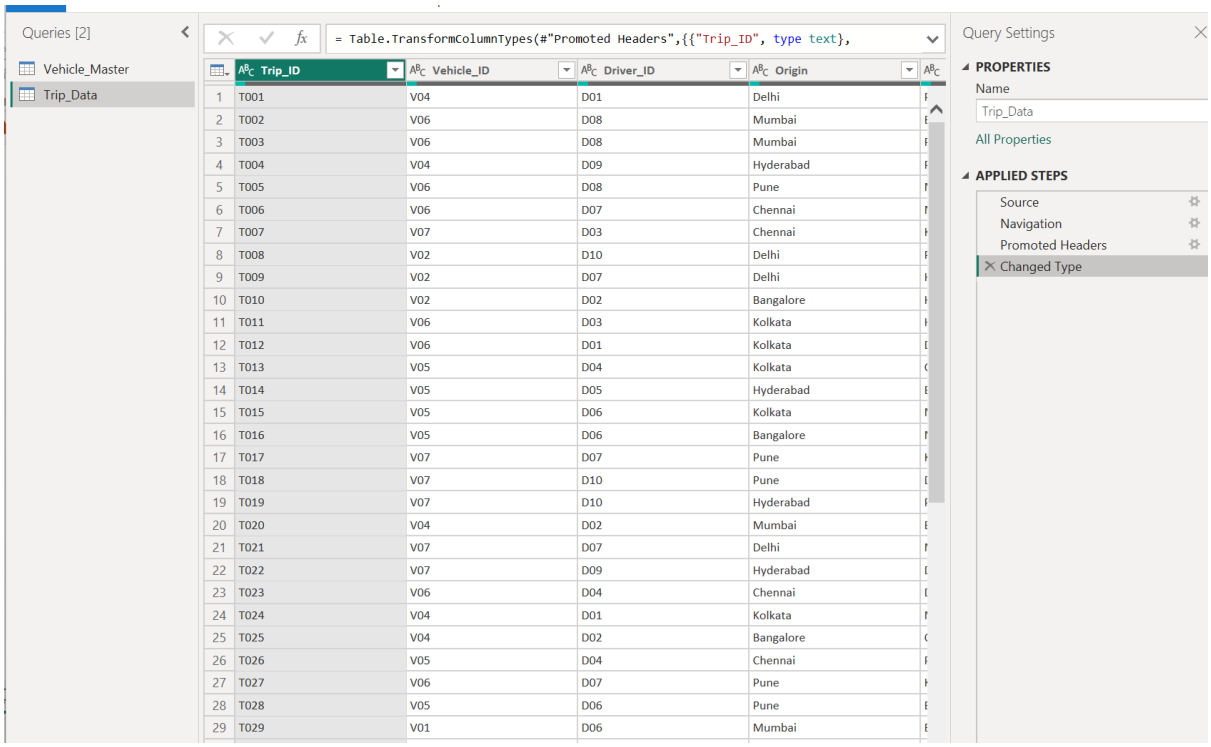
1. Data Cleaning & Modeling:

○ Fix missing fuel consumption values (use mean imputation).

Step1: Import Data into PowerBI

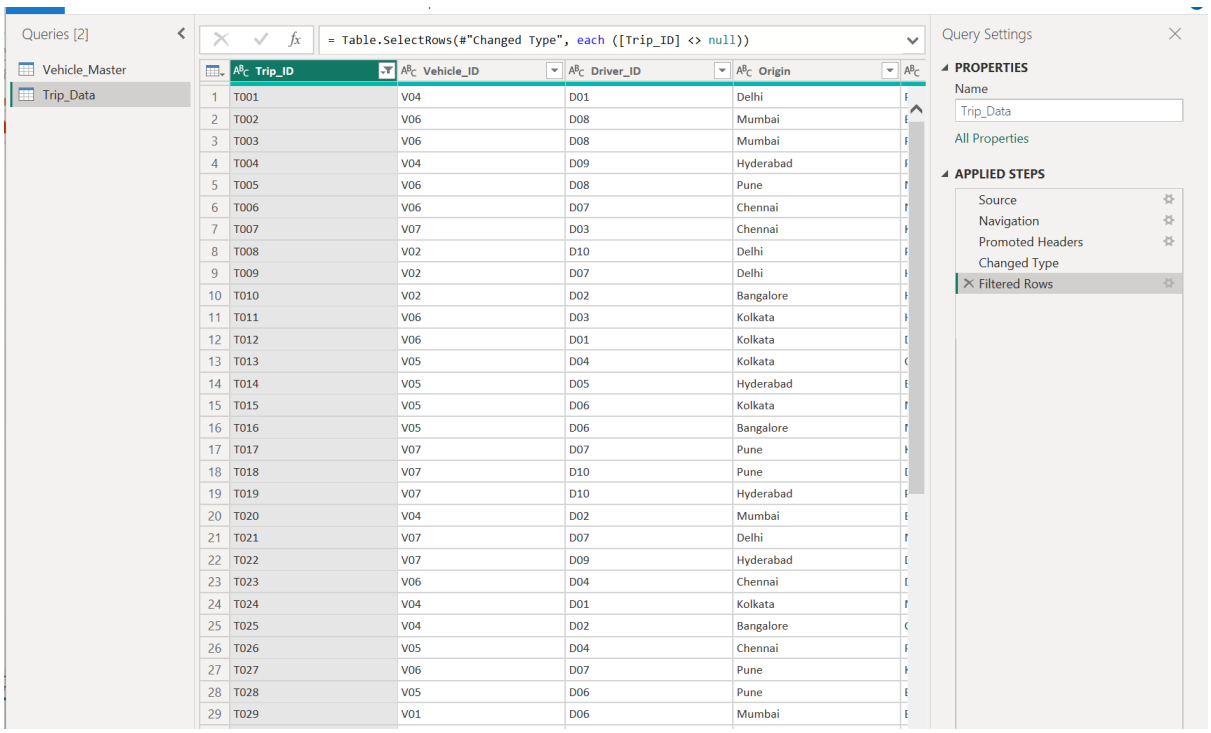


Step2: Transform Data into Power Query



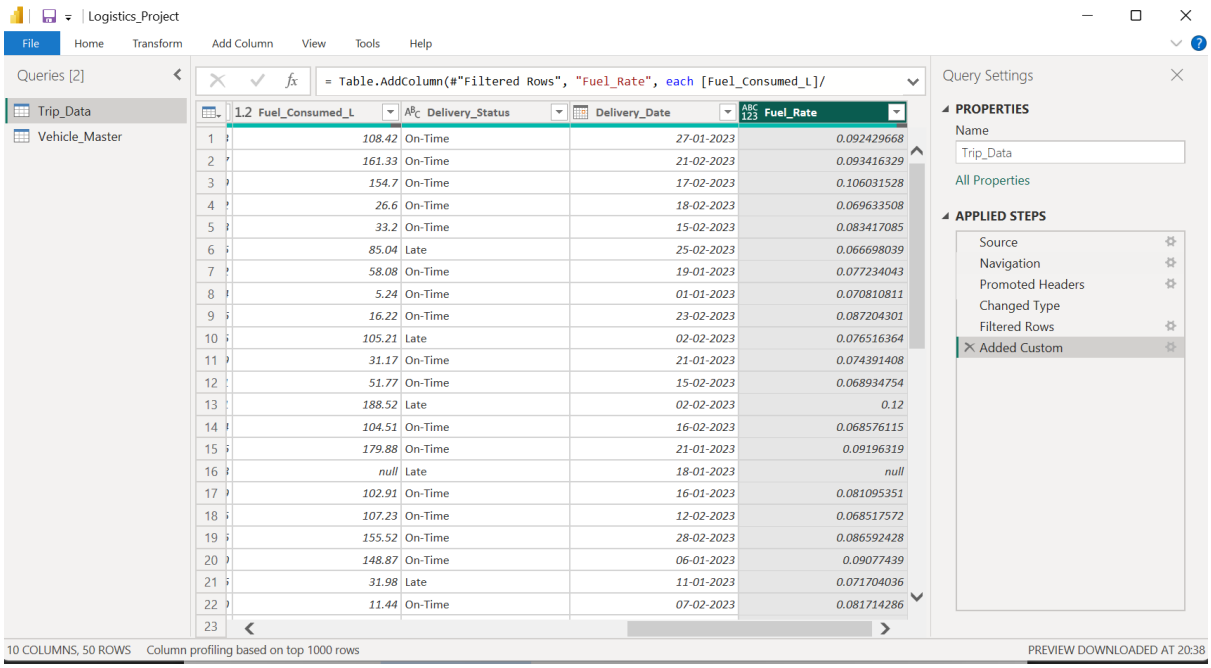
	Trip_ID	Vehicle_ID	Driver_ID	Origin
1	T001	V04	D01	Delhi
2	T002	V06	D08	Mumbai
3	T003	V06	D08	Mumbai
4	T004	V04	D09	Hyderabad
5	T005	V06	D08	Pune
6	T006	V06	D07	Chennai
7	T007	V07	D03	Chennai
8	T008	V02	D10	Delhi
9	T009	V02	D07	Delhi
10	T010	V02	D02	Bangalore
11	T011	V06	D03	Kolkata
12	T012	V06	D01	Kolkata
13	T013	V05	D04	Kolkata
14	T014	V05	D05	Hyderabad
15	T015	V05	D06	Kolkata
16	T016	V05	D06	Bangalore
17	T017	V07	D07	Pune
18	T018	V07	D10	Pune
19	T019	V07	D10	Hyderabad
20	T020	V04	D02	Mumbai
21	T021	V07	D07	Delhi
22	T022	V07	D09	Hyderabad
23	T023	V06	D04	Chennai
24	T024	V04	D01	Kolkata
25	T025	V04	D02	Bangalore
26	T026	V05	D04	Chennai
27	T027	V06	D07	Pune
28	T028	V05	D06	Pune
29	T029	V01	D06	Mumbai

Step3: unchecked nulls in Trip_Id



	Trip_ID	Vehicle_ID	Driver_ID	Origin
1	T001	V04	D01	Delhi
2	T002	V06	D08	Mumbai
3	T003	V06	D08	Mumbai
4	T004	V04	D09	Hyderabad
5	T005	V06	D08	Pune
6	T006	V06	D07	Chennai
7	T007	V07	D03	Chennai
8	T008	V02	D10	Delhi
9	T009	V02	D07	Delhi
10	T010	V02	D02	Bangalore
11	T011	V06	D03	Kolkata
12	T012	V06	D01	Kolkata
13	T013	V05	D04	Kolkata
14	T014	V05	D05	Hyderabad
15	T015	V05	D06	Kolkata
16	T016	V05	D06	Bangalore
17	T017	V07	D07	Pune
18	T018	V07	D10	Pune
19	T019	V07	D10	Hyderabad
20	T020	V04	D02	Mumbai
21	T021	V07	D07	Delhi
22	T022	V07	D09	Hyderabad
23	T023	V06	D04	Chennai
24	T024	V04	D01	Kolkata
25	T025	V04	D02	Bangalore
26	T026	V05	D04	Chennai
27	T027	V06	D07	Pune
28	T028	V05	D06	Pune
29	T029	V01	D06	Mumbai

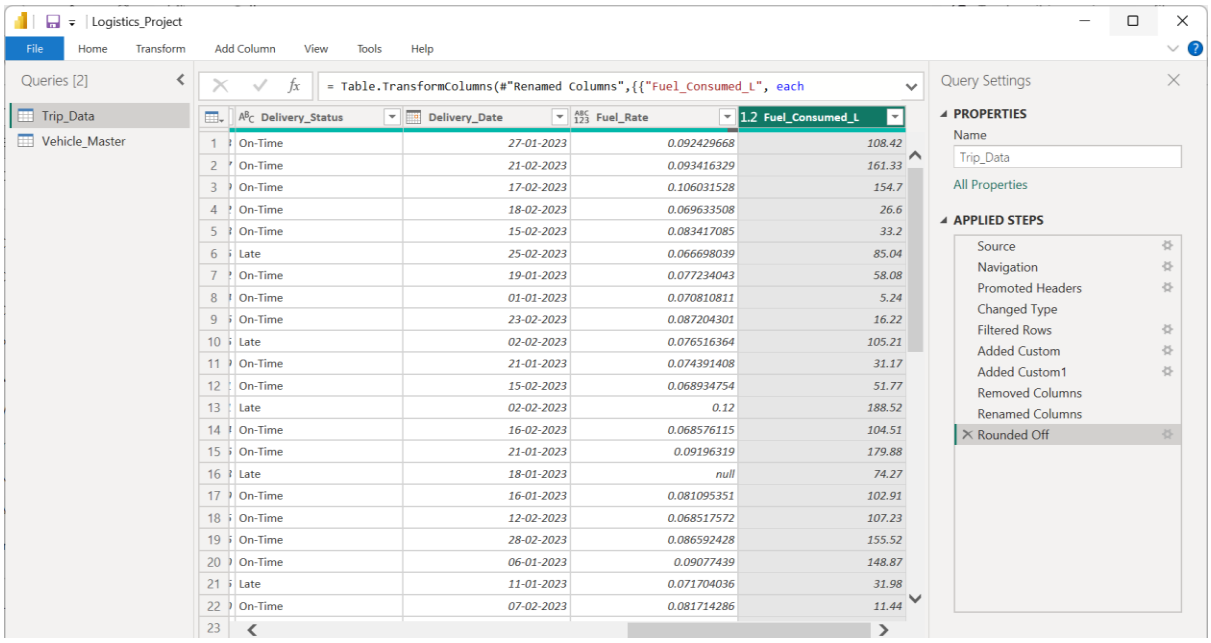
Step4: A new custom column (Fuel_Rate) was created in Power Query by dividing Fuel_Consumed_L by Distance_km to determine fuel consumption per kilometer.



	1.2 Fuel_Consumed_L	Delivery_Status	Delivery_Date	Fuel_Rate
1	108.42	On-Time	27-01-2023	0.092429668
2	161.33	On-Time	21-02-2023	0.093416329
3	154.7	On-Time	17-02-2023	0.106031528
4	26.6	On-Time	18-02-2023	0.069633508
5	33.2	On-Time	15-02-2023	0.083417085
6	85.04	Late	25-02-2023	0.066698039
7	58.08	On-Time	19-01-2023	0.077234043
8	5.24	On-Time	01-01-2023	0.070810811
9	16.22	On-Time	23-02-2023	0.087204301
10	105.21	Late	02-02-2023	0.076516364
11	31.17	On-Time	21-01-2023	0.074391408
12	51.77	On-Time	15-02-2023	0.068934754
13	188.52	Late	02-02-2023	0.12
14	104.51	On-Time	16-02-2023	0.068576115
15	179.88	On-Time	21-01-2023	0.09196319
16	null	Late	18-01-2023	null
17	102.91	On-Time	16-01-2023	0.081095351
18	107.23	On-Time	12-02-2023	0.068517572
19	155.52	On-Time	28-02-2023	0.086592428
20	148.87	On-Time	06-01-2023	0.09077439
21	31.98	Late	11-01-2023	0.071704036
22	11.44	On-Time	07-02-2023	0.081714286
23				

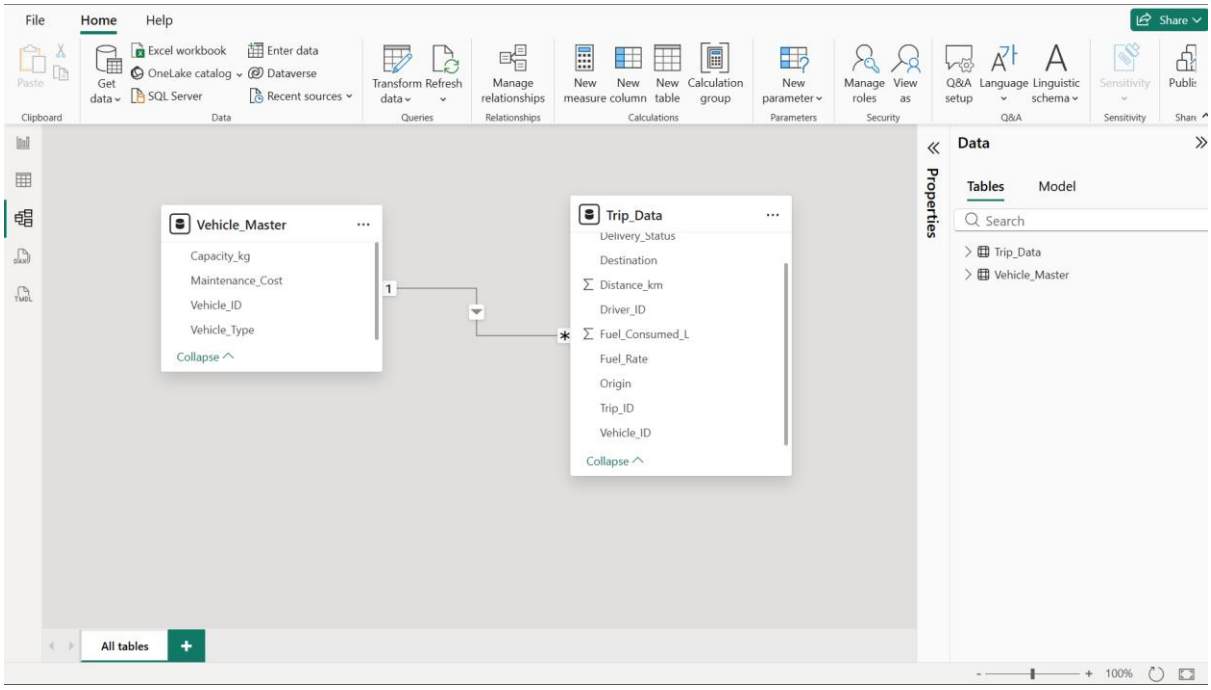
Step5: The average fuel rate was calculated using the Statistics → Average function in Power Query to support mean imputation.

Step6: Filled null values by using mean imputation



	Delivery_Status	Delivery_Date	Fuel_Rate	Fuel_Consumed_L
1	On-Time	27-01-2023	0.092429668	108.42
2	On-Time	21-02-2023	0.093416329	161.33
3	On-Time	17-02-2023	0.106031528	154.7
4	On-Time	18-02-2023	0.069633508	26.6
5	On-Time	15-02-2023	0.083417085	33.2
6	Late	25-02-2023	0.066698039	85.04
7	On-Time	19-01-2023	0.077234043	58.08
8	On-Time	01-01-2023	0.070810811	5.24
9	On-Time	23-02-2023	0.087204301	16.22
10	Late	02-02-2023	0.076516364	105.21
11	On-Time	21-01-2023	0.074391408	31.17
12	On-Time	15-02-2023	0.068934754	51.77
13	Late	02-02-2023	0.12	188.52
14	On-Time	16-02-2023	0.068576115	104.51
15	On-Time	21-01-2023	0.09196319	179.88
16	Late	18-01-2023	null	74.27
17	On-Time	16-01-2023	0.081095351	102.91
18	On-Time	12-02-2023	0.068517572	107.23
19	On-Time	28-02-2023	0.086592428	155.52
20	On-Time	06-01-2023	0.09077439	148.87
21	Late	11-01-2023	0.071704036	31.98
22	On-Time	07-02-2023	0.081714286	11.44
23				

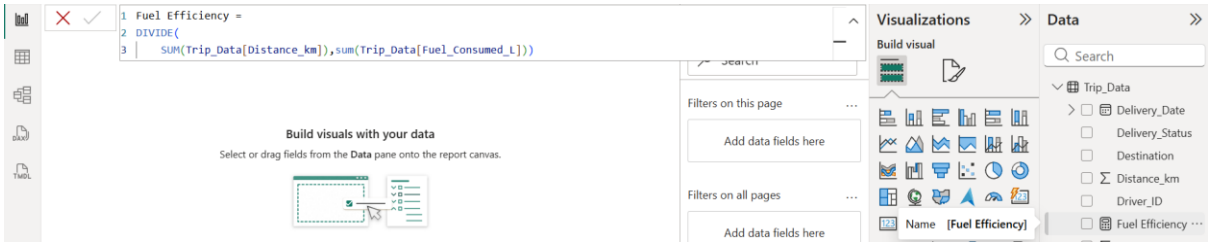
Step7: Create a relationship using Vehicle_ID between Trip_Data and the Vehicle Master table.



A one to many relationship was established between Trip_Data and Vehicle_Master using Vehicle_ID as the primary key with single cross-filter direction.

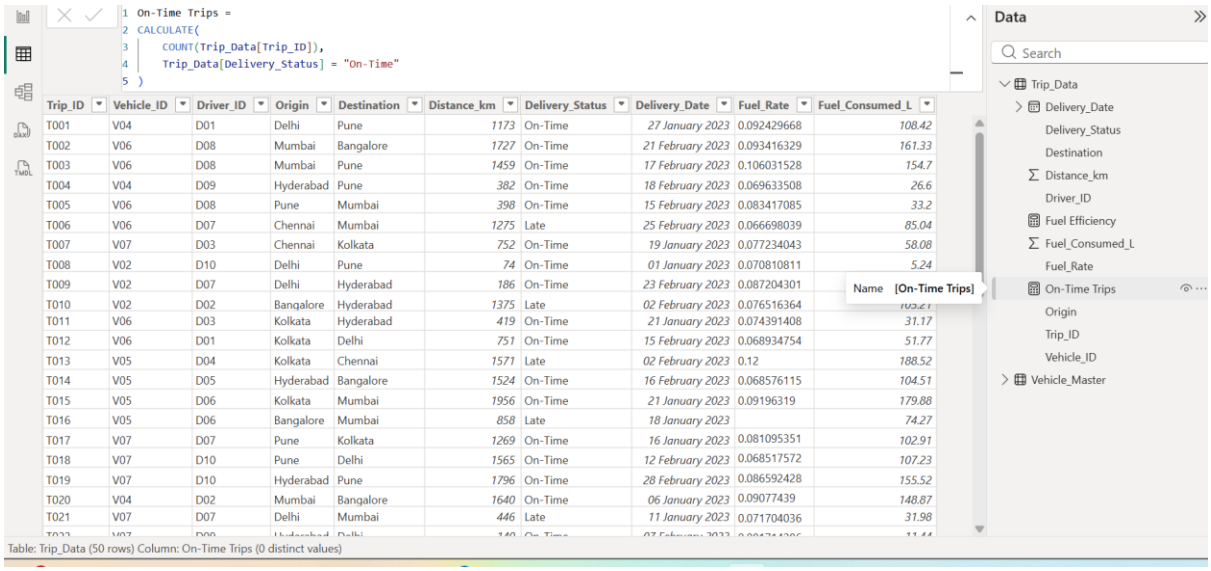
2. DAX Measures:

Step1: To Calculate Fuel Efficiency



Fuel Efficiency was calculated as Total Distance divided by Total Fuel Consumed using DAX.

Step2: To Calculate On time trips



```
On-Time Trips =
CALCULATE(
    COUNT(Trip_Data[Trip_ID]),
    Trip_Data[Delivery_Status] = "On-Time"
)
```

Trip_ID	Vehicle_ID	Driver_ID	Origin	Destination	Distance_km	Delivery_Status	Delivery_Date	Fuel_Rate	Fuel_Consumed_L
T001	V04	D01	Delhi	Pune	1173	On-Time	27 January 2023	0.092429668	108.42
T002	V06	D08	Mumbai	Bangalore	1727	On-Time	21 February 2023	0.093416329	161.33
T003	V06	D08	Mumbai	Pune	1459	On-Time	17 February 2023	0.106031528	154.7
T004	V04	D09	Hyderabad	Pune	382	On-Time	18 February 2023	0.069633508	26.6
T005	V06	D08	Pune	Mumbai	398	On-Time	15 February 2023	0.083417085	33.2
T006	V06	D07	Chennai	Mumbai	1275	Late	25 February 2023	0.066698039	85.04
T007	V07	D03	Chennai	Kolkata	752	On-Time	19 January 2023	0.077234043	58.08
T008	V02	D10	Delhi	Pune	74	On-Time	01 January 2023	0.070810811	5.24
T009	V02	D07	Delhi	Hyderabad	186	On-Time	23 February 2023	0.087204301	
T010	V02	D02	Bangalore	Hyderabad	1375	Late	02 February 2023	0.076516364	
T011	V06	D03	Kolkata	Hyderabad	419	On-Time	21 January 2023	0.074391408	31.17
T012	V06	D01	Kolkata	Delhi	751	On-Time	15 February 2023	0.068934754	51.77
T013	V05	D04	Kolkata	Chennai	1571	Late	02 February 2023	0.12	188.52
T014	V05	D05	Hyderabad	Bangalore	1524	On-Time	16 February 2023	0.068576115	104.51
T015	V05	D06	Kolkata	Mumbai	1956	On-Time	21 January 2023	0.09196319	179.88
T016	V05	D06	Bangalore	Mumbai	858	Late	18 January 2023		74.27
T017	V07	D07	Pune	Kolkata	1269	On-Time	16 January 2023	0.081095351	102.91
T018	V07	D10	Pune	Delhi	1565	On-Time	12 February 2023	0.068517572	107.23
T019	V07	D10	Hyderabad	Pune	1796	On-Time	28 February 2023	0.086592428	155.52
T020	V04	D02	Mumbai	Bangalore	1640	On-Time	06 January 2023	0.09077439	148.87
T021	V07	D07	Delhi	Mumbai	446	Late	11 January 2023	0.071704036	31.98

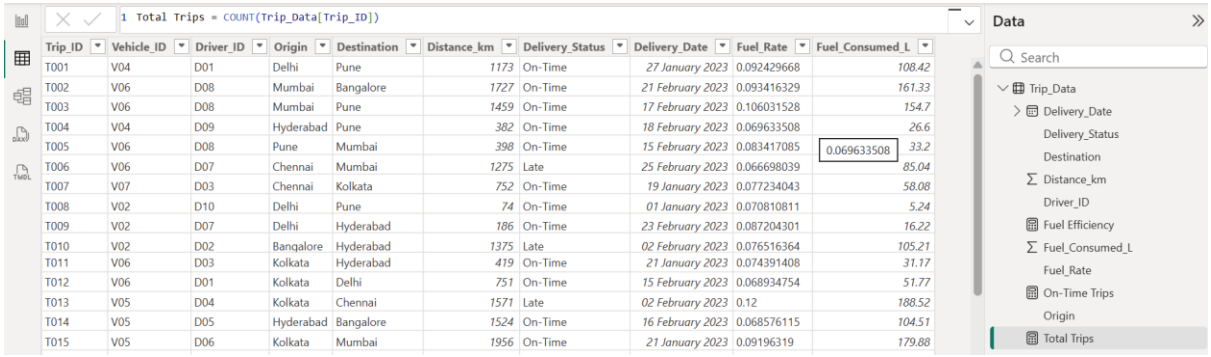
Table: Trip_Data (50 rows) Column: On-Time Trips (0 distinct values)

Name [On-Time Trips]

Data: Search; Trip_Data; Delivery_Date; Delivery_Status; Destination; Distance_km; Driver_ID; Fuel Efficiency; Fuel_Consumed_L; Fuel_Rate; On-Time Trips; Origin; Trip_ID; Vehicle_ID; Vehicle_Master

Ontime trips calculated by counting tripid and delivery status

Step3: Total trips



```
Total Trips = COUNT(Trip_Data[Trip_ID])
```

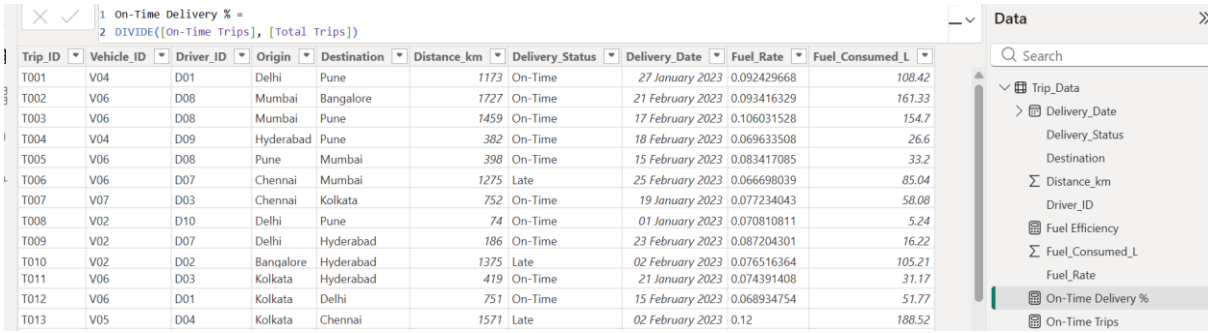
Trip_ID	Vehicle_ID	Driver_ID	Origin	Destination	Distance_km	Delivery_Status	Delivery_Date	Fuel_Rate	Fuel_Consumed_L
T001	V04	D01	Delhi	Pune	1173	On-Time	27 January 2023	0.092429668	108.42
T002	V06	D08	Mumbai	Bangalore	1727	On-Time	21 February 2023	0.093416329	161.33
T003	V06	D08	Mumbai	Pune	1459	On-Time	17 February 2023	0.106031528	154.7
T004	V04	D09	Hyderabad	Pune	382	On-Time	18 February 2023	0.069633508	26.6
T005	V06	D08	Pune	Mumbai	398	On-Time	15 February 2023	0.083417085	33.2
T006	V06	D07	Chennai	Mumbai	1275	Late	25 February 2023	0.066698039	85.04
T007	V07	D03	Chennai	Kolkata	752	On-Time	19 January 2023	0.077234043	58.08
T008	V02	D10	Delhi	Pune	74	On-Time	01 January 2023	0.070810811	5.24
T009	V02	D07	Delhi	Hyderabad	186	On-Time	23 February 2023	0.087204301	16.22
T010	V02	D02	Bangalore	Hyderabad	1375	Late	02 February 2023	0.076516364	105.21
T011	V06	D03	Kolkata	Hyderabad	419	On-Time	21 January 2023	0.074391408	31.17
T012	V06	D01	Kolkata	Delhi	751	On-Time	15 February 2023	0.068934754	51.77
T013	V05	D04	Kolkata	Chennai	1571	Late	02 February 2023	0.12	188.52
T014	V05	D05	Hyderabad	Bangalore	1524	On-Time	16 February 2023	0.068576115	104.51
T015	V05	D06	Kolkata	Mumbai	1956	On-Time	21 January 2023	0.09196319	179.88

0.069633508

Data: Search; Trip_Data; Delivery_Date; Delivery_Status; Destination; Distance_km; Driver_ID; Fuel Efficiency; Fuel_Consumed_L; Fuel_Rate; On-Time Trips; Origin; Total Trips

Total trips calculated by counting Trip_Data Coulmn using DAX measures.

Step4: On-Time Delivery %



```
On-Time Delivery % =
DIVIDE([On-Time Trips], [Total Trips])
```

Trip_ID	Vehicle_ID	Driver_ID	Origin	Destination	Distance_km	Delivery_Status	Delivery_Date	Fuel_Rate	Fuel_Consumed_L
T001	V04	D01	Delhi	Pune	1173	On-Time	27 January 2023	0.092429668	108.42
T002	V06	D08	Mumbai	Bangalore	1727	On-Time	21 February 2023	0.093416329	161.33
T003	V06	D08	Mumbai	Pune	1459	On-Time	17 February 2023	0.106031528	154.7
T004	V04	D09	Hyderabad	Pune	382	On-Time	18 February 2023	0.069633508	26.6
T005	V06	D08	Pune	Mumbai	398	On-Time	15 February 2023	0.083417085	33.2
T006	V06	D07	Chennai	Mumbai	1275	Late	25 February 2023	0.066698039	85.04
T007	V07	D03	Chennai	Kolkata	752	On-Time	19 January 2023	0.077234043	58.08
T008	V02	D10	Delhi	Pune	74	On-Time	01 January 2023	0.070810811	5.24
T009	V02	D07	Delhi	Hyderabad	186	On-Time	23 February 2023	0.087204301	16.22
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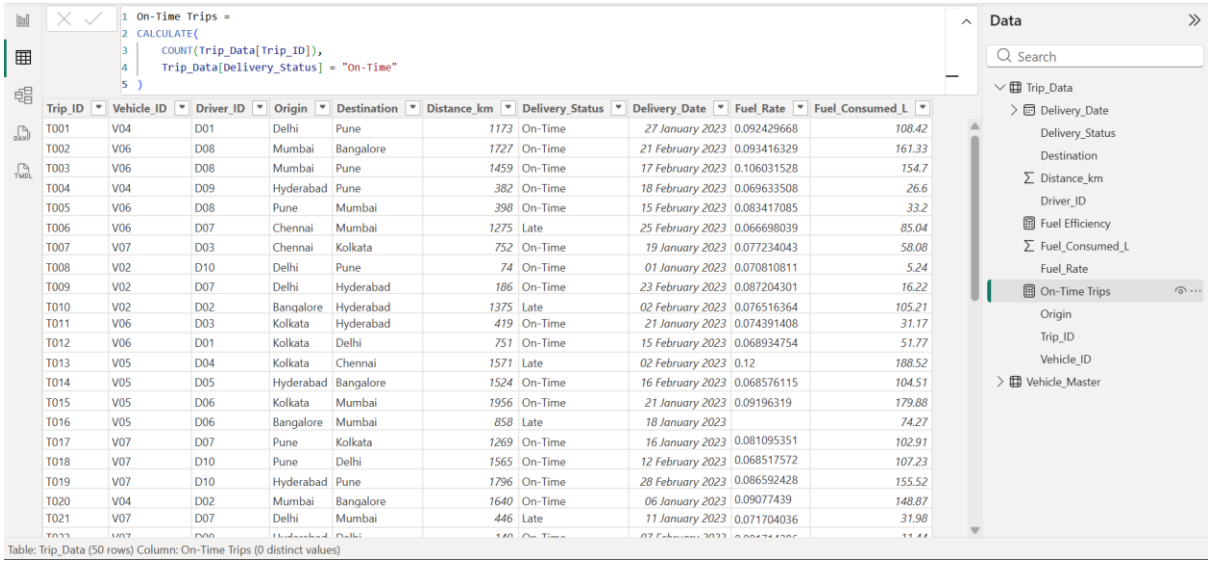
T011	V06	D03	Kolkata	Hyderabad	419	On-Time	21 January 2023	0.074391408	31.17
T012	V06	D01	Kolkata	Delhi	751	On-Time	15 February 2023	0.068934754	51.77
T013	V05	D04	Kolkata	Chennai	1571	Late	02 February 2023	0.12	188.52

Data: Search; Trip_Data; Delivery_Date; Delivery_Status; Destination; Distance_km; Driver_ID; Fuel Efficiency; Fuel_Consumed_L; Fuel_Rate; On-Time Delivery %; On-Time Trips

On-Time Delivery percentage was calculated by dividing the number of On-Time trips by total trips using DAX measures.

Step5: On-Time Delivery % = On-Time Trips / Total Trips:

DAX measure to find On-Time Trips



```
On-Time Trips =
CALCULATE(
    COUNT(Trip_Data[Trip_ID]),
    Trip_Data[Delivery_Status] = "On-Time"
)
```

Trip_ID	Vehicle_ID	Driver_ID	Origin	Destination	Distance_km	Delivery_Status	Delivery_Date	Fuel_Rate	Fuel_Consumed_L
T001	V04	D01	Delhi	Pune	1173	On-Time	27 January 2023	0.092429668	108.42
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T003	V06	D08	Mumbai	Pune	1459	On-Time	17 February 2023	0.106031528	154.7
T004	V04	D09	Hyderabad	Pune	382	On-Time	18 February 2023	0.069633508	26.6
T005	V06	D08	Pune	Mumbai	398	On-Time	15 February 2023	0.083417085	33.2
T006	V06	D07	Chennai	Mumbai	1275	Late	25 February 2023	0.066698039	85.04
T007	V07	D03	Chennai	Kolkata	752	On-Time	19 January 2023	0.077234043	58.08
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T009	V02	D07	Delhi	Hyderabad	186	On-Time	23 February 2023	0.087204301	16.22
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T013	V05	D04	Kolkata	Chennai	1571	Late	02 February 2023	0.12	188.52
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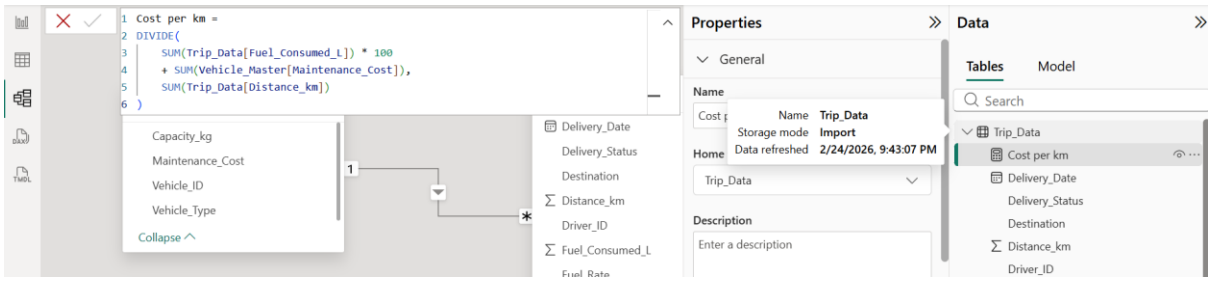
T015	V05	D06	Kolkata	Mumbai	1956	On-Time	21 January 2023	0.09196319	179.88
T016	V05	D06	Bangalore	Mumbai	858	Late	18 January 2023		74.27
T017	V07	D07	Pune	Kolkata	1269	On-Time	16 January 2023	0.081095351	102.91
T018	V07	D10	Pune	Delhi	1565	On-Time	12 February 2023	0.068517572	107.23
T019	V07	D10	Hyderabad	Pune	1796	On-Time	28 February 2023	0.086592428	155.52
T020	V04	D02	Mumbai	Bangalore	1640	On-Time	06 January 2023	0.09077439	148.87
T021	V07	D07	Delhi	Mumbai	446	Late	11 January 2023	0.071704036	31.98

Cost per km was calculated as (Fuel Consumed × 100 + Maintenance Cost) divided by total distance using DAX.

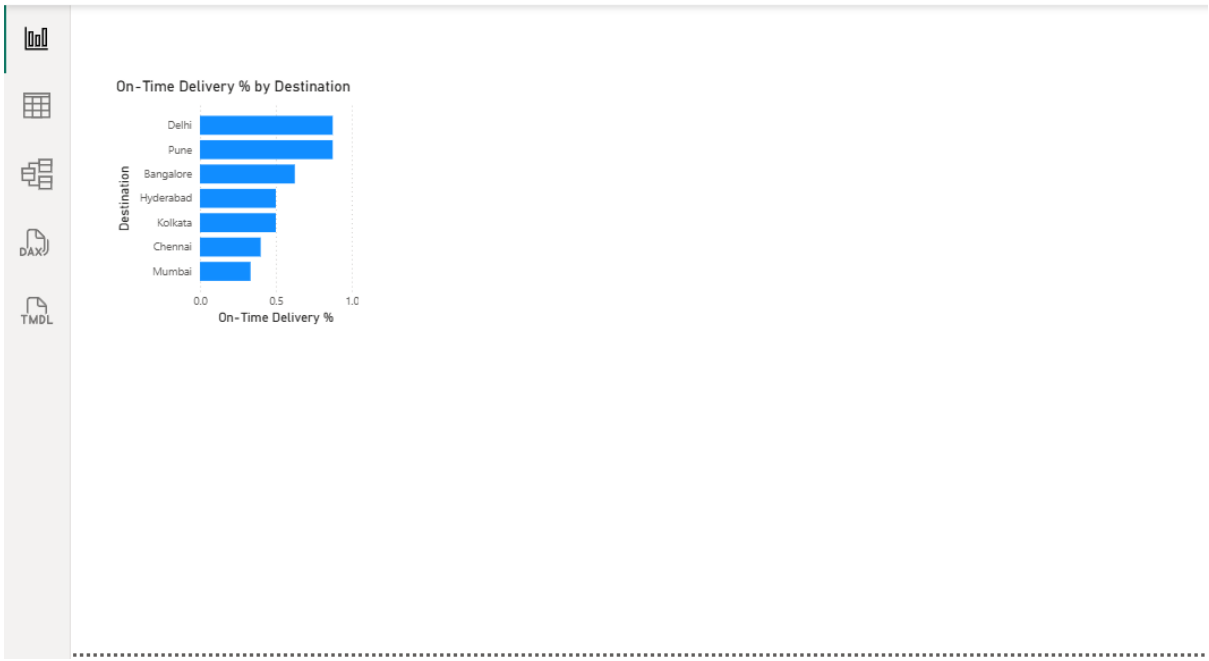
Step6: Cost per km = (fuel cost + Maintenance Cost) / Distance



```
Cost per km =
DIVIDE(
    SUM(Trip_Data[Fuel_Consumed_L]) * 100
    + SUM(Vehicle_Master[Maintenance_Cost]),
    SUM(Trip_Data[Distance_km])
)
```

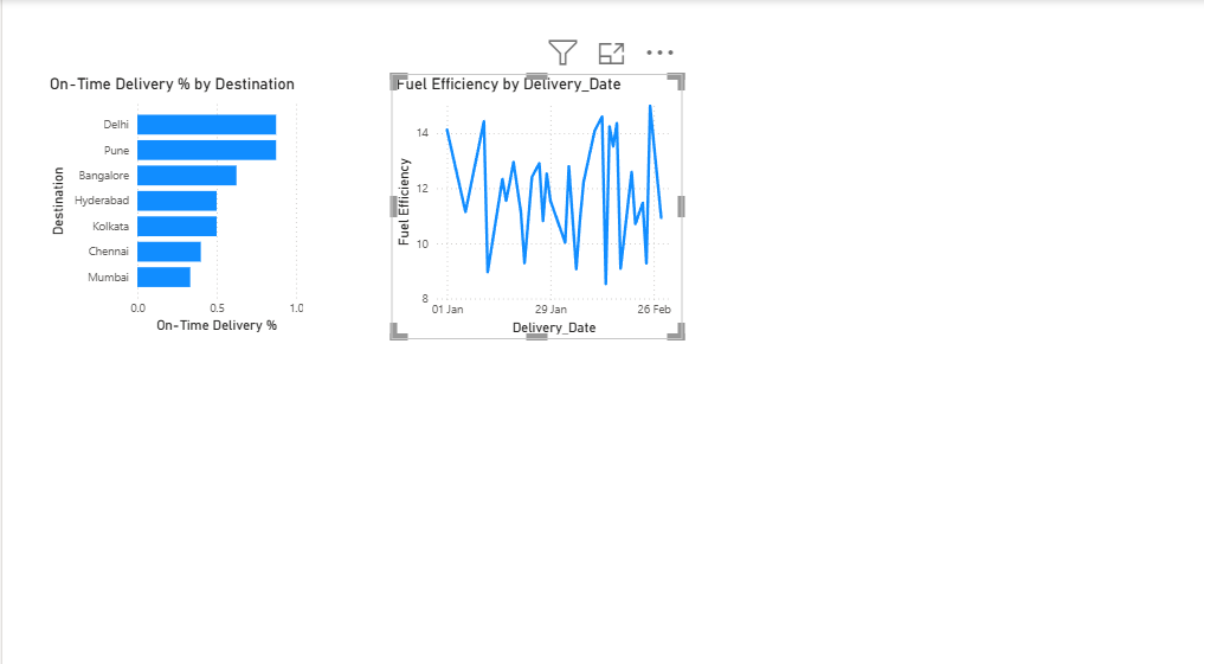
3. Visualization:

Bar chart: On-Time Delivery % by Destination.



A clustered bar chart was created to analyze On-Time Delivery percentage across different destinations.

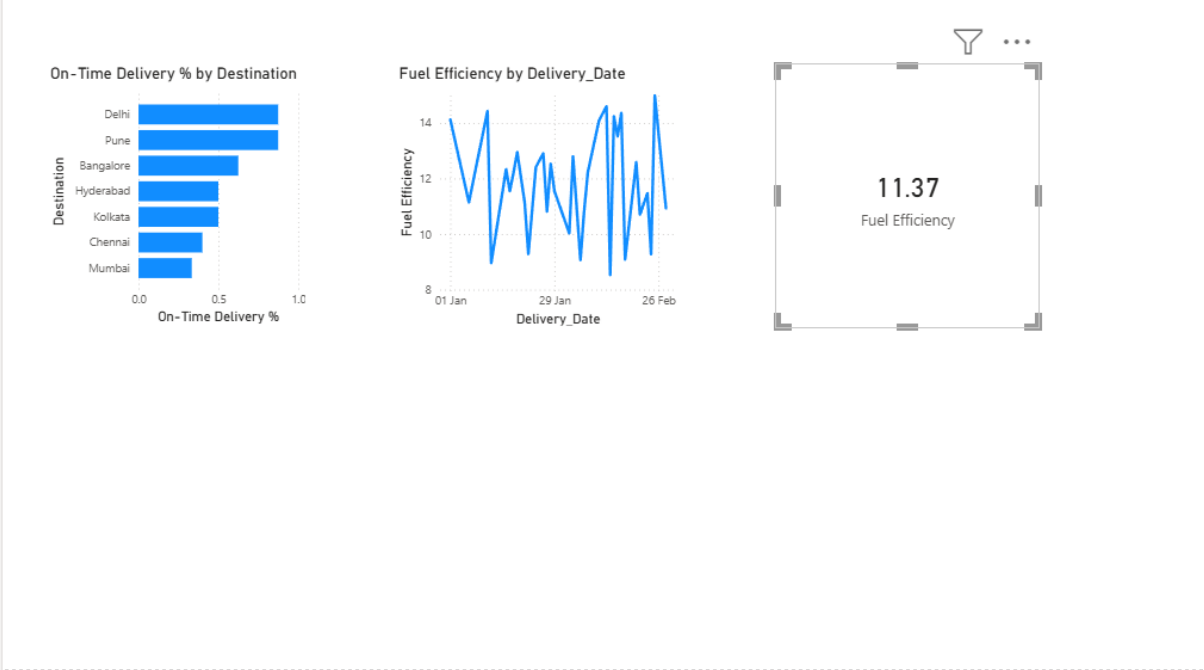
Line chart: Fuel efficiency trend by delivery date.



A line chart was created to analyze the trend of fuel efficiency across delivery dates

Cards visualization:

Step1: Fuel Efficiency

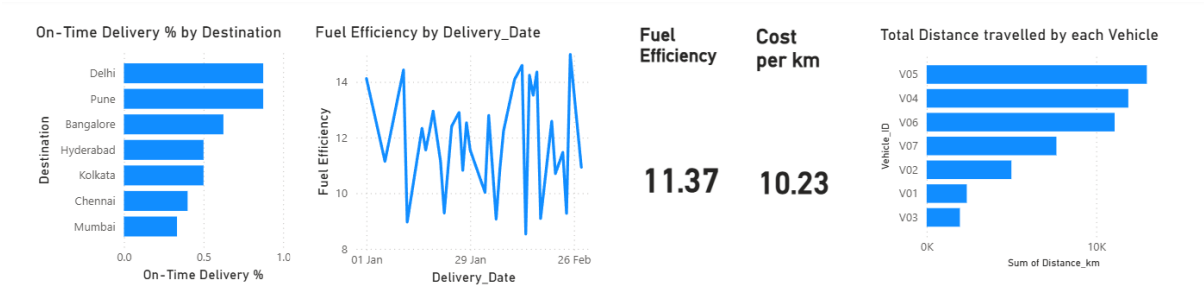


A card visual was created to display overall Fuel Efficiency, calculated as total distance divided by total fuel consumed.

Step2: Average Cost per km

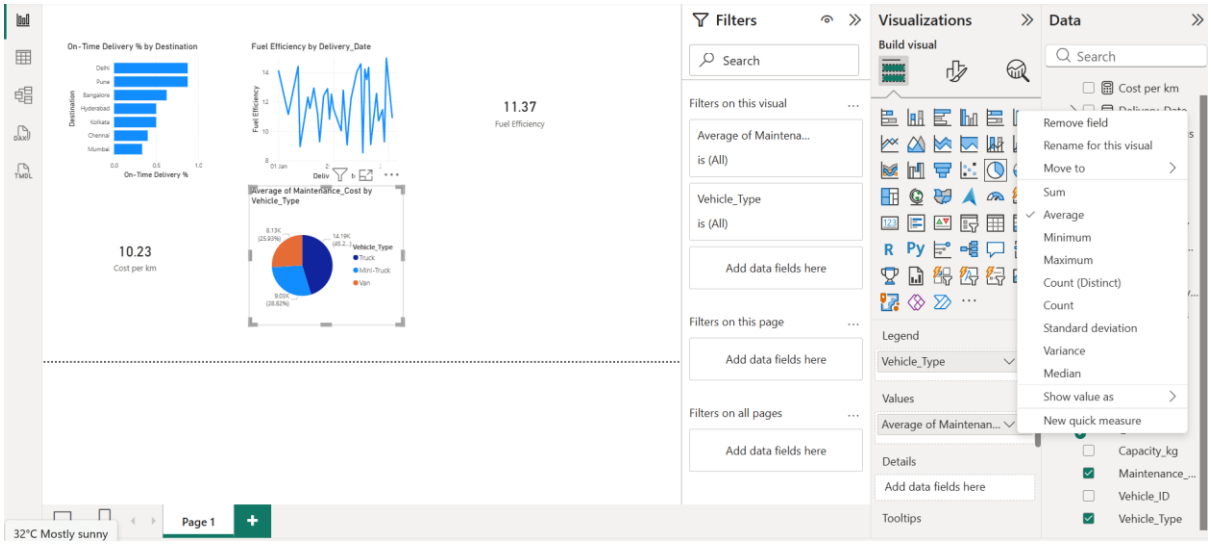
A card visual was created to display Average Cost per km, calculated as total operational cost divided by total distance traveled.

Step3: Total distance travelled by each vehicle



A clustered bar chart was created to display the total distance travelled by each vehicle.

Pie chart: vehicle type vs Average maintenance cost



A pie chart was created to analyze the distribution of average maintenance cost across different vehicle types.

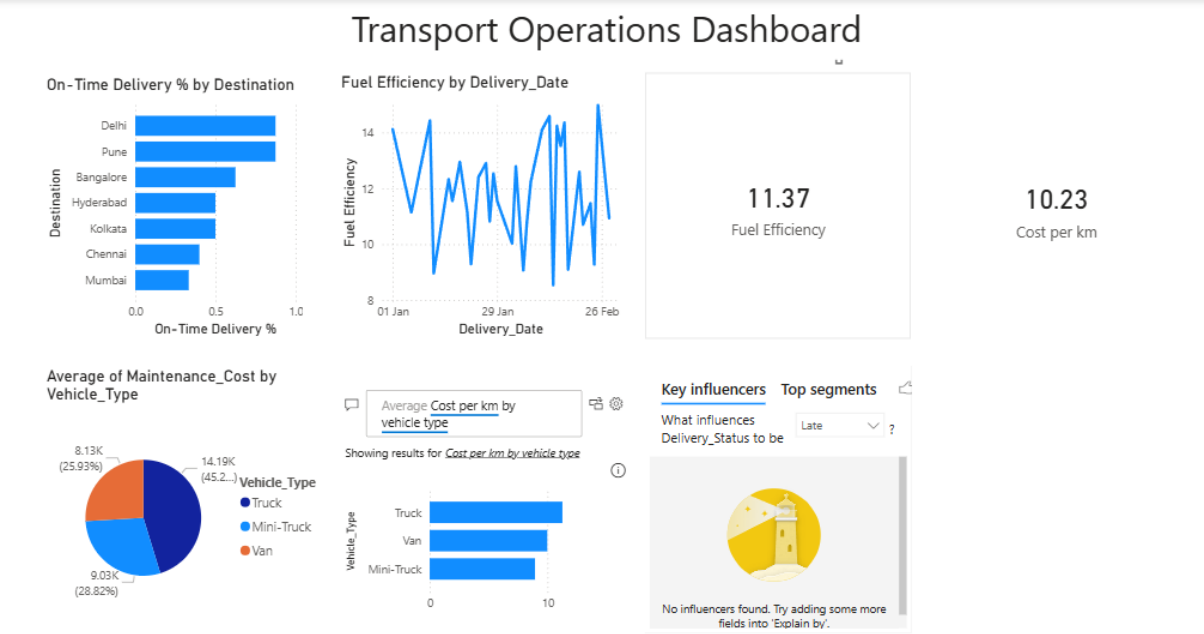
4. AI-Powered Visuals

Q&A Visual:

The Q&A visual was used to generate insights dynamically by querying “Average Cost per km by vehicle type.

Key Influencers Visual:

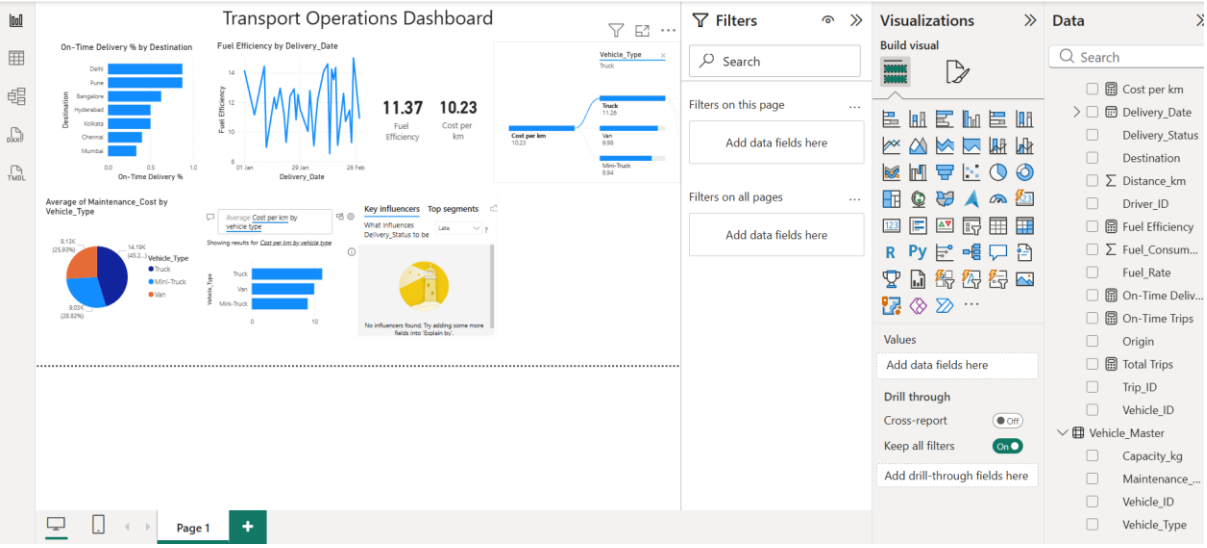
➢ Add Delivery Status in the Analyze and explain by -distance in km,vehicle_type, Driver_ID



The Key Influencers visual was used to analyze factors affecting Delivery Status based on Distance_km, Vehicle_Type, and Driver_ID

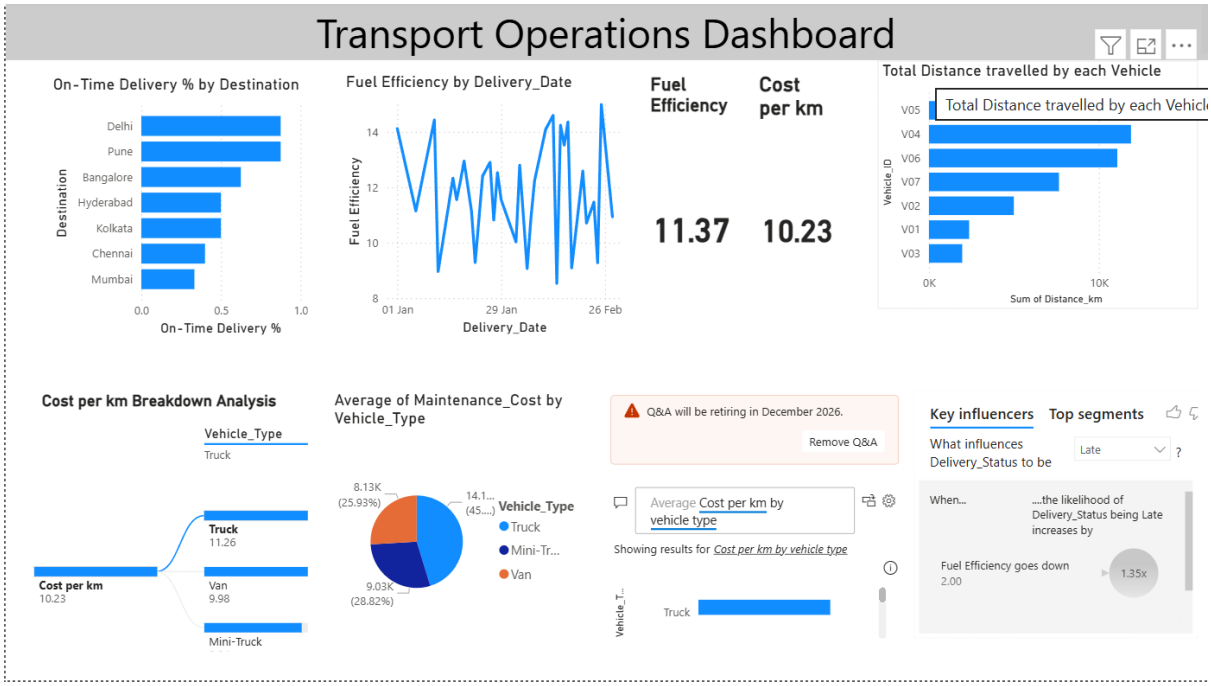
Decomposition Tree (AI Visual):

➢ Analyze Cost per km → explained by Vehicle Type, Maintenance_cost, and Distance_km.



A Decomposition Tree was created to analyze Cost per km and break it down by Vehicle Type. The visual allows further exploration using Maintenance_Cost and Distance_km as explanatory factors.

The final Transport Operations Dashboard integrates performance metrics, trend analysis, cost evaluation, and AI-driven insights to support operational optimization.



Conclusion

The Transport Operations Dashboard was developed using Power BI to analyze delivery performance, fuel efficiency, and operational cost. Data cleaning and modeling were performed to ensure accuracy. DAX measures were created to calculate key metrics such as Fuel Efficiency, On-Time Delivery %, and Cost per km. The dashboard integrates visual and AI-driven insights to support data-driven decision-making and fleet optimization